Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2026.DOI

Temporal Pixel Masking for Mitigating OCR-Based Text Recovery from Short-Exposure Screen Photography: A Feasibility Study with a UVC Camera

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[PLACEHOLDER: Funding acknowledgment to be filled.] This manuscript includes a minimal-risk human-subject usability study. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type. Before each session, the experimenter individually confirmed informed consent and photosensitivity-screening eligibility in person. The study also followed the withdrawal safeguards described in Section V-E, and only de-identified data are analyzed and reported.

ABSTRACT Screen content can be covertly photographed and transcribed by optical character recognition (OCR) or vision-language models (VLMs). We present temporal pixel masking, a display method exploiting differences between human temporal integration and short camera exposures. A base frame is decomposed into rapidly alternating, cryptographically randomized complementary subframes. We evaluate a 240-Hz software prototype using one USB Video Class (UVC) webcam. In the primary matched short-exposure analysis (12 content items $\times$ 9 geometries $\times$ 3 repeats), best-of-engine OCR character recovery decreased from 94.1% unprotected to 16.2% for the readability-priority profile and 5.0% for the high-suppression profile; paired content-cluster contrasts were 78.0 and 89.1 percentage points, respectively. The latter meets the descriptive $<$10% recovery target. Across the same nine geometries, 150-frame temporal averaging recovered 47.9–71.1% of characters. In the 0.5 m VLM probe, the strongest of three tested VLMs reached 77.8% exact match, with recovery concentrated on large-font short snippets. Under long exposure, masking could increase recovery over the unprotected baseline, consistent with reduced luminance shifting the sensor into its linear range. These results establish feasibility for mitigating conventional OCR in the tested short-exposure UVC-camera link, but not protection against temporal integration or VLM attackers. In 56 participants, readability-priority masking retained 94.83% typing accuracy but reduced speed by 1.14 words per minute (WPM); readability, stability, and comfort averaged 3.79, 2.98, and 3.39 out of 5, indicating measurable usability costs.

INDEX TERMS Adversarial noise, camera capture, display privacy, optical character recognition, persistence of vision, temporal masking, vision-language models, visual eavesdropping

I. INTRODUCTION

SCREENS routinely display credentials, financial records, source code, and other sensitive information in offices, banks, hospitals, and shared workspaces. At the same time, smartphones, smart glasses [1], and miniature cameras make physical screen capture increasingly unobtrusive. A single photograph can be processed by optical character recognition (OCR) or a vision-language model (VLM) without accessing the host device or leaving a network trace. Controlled experiments and user studies further show that visual eavesdropping can succeed quickly and often goes unnoticed [2], [3].

Existing defenses address different parts of this threat. Physical privacy filters and spatial display transformations restrict viewing geometry, while digital rights management and watermarking target digital copying or post-capture attribution [4], [5]. Temporal display techniques have also used persistence of vision and psychovisual modulation to separate what viewers and cameras perceive [6]–[10]. These studies establish temporal modulation as a viable display mechanism, but they leave an important empirical question unresolved. It remains unclear how much sensitive text is machine-recoverable from physical short-exposure captures

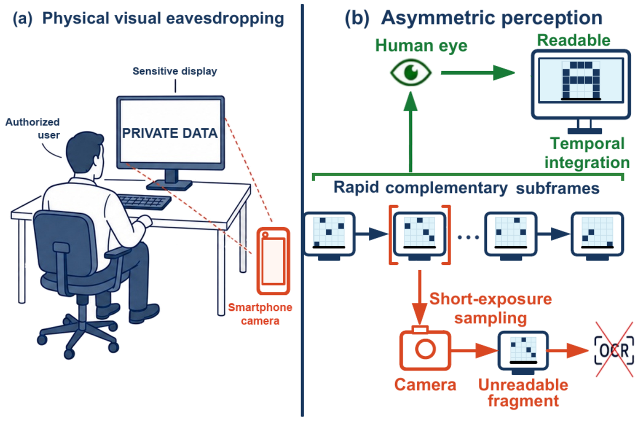

FIGURE 1. Threat scenario and operating principle of temporal pixel masking. (a) A nearby camera can capture sensitive on-screen content without accessing the host system. (b) Human vision integrates complementary subframes, while a short camera exposure can intersect only part of a display cycle and leave OCR with an incomplete fragment.

and how rapidly recovery increases with stronger recognition and temporal reconstruction methods.

We investigate this question through *temporal pixel masking*, a display-side implementation that builds on prior temporal masking principles. Each frame is decomposed into n rapidly alternating subframes whose pixel assignments are randomized, mutually exclusive, and collectively complete in the ideal digital domain. A human observer can integrate information across the display cycle, whereas a short camera exposure may intersect only part of that cycle and record incomplete character structure (Fig. 1). Profile-dependent stripe and glyph overlays, complementary noise, and a partial inversion frame provide additional operating points between readability and suppression. This mechanism does not prevent reconstruction in general: a long exposure or registered multi-frame sequence can accumulate the complementary information.

We implement the method as a 240-Hz graphics processing unit (GPU) rendering prototype and evaluate it with an instrumented USB Video Class (UVC) capture pipeline. The physical archive contains 10,575 captures from one webcam across three distances and three viewing angles. In the primary matched analysis at the common nominal exposure setting, conventional OCR character recovery decreases from 94.1% for the unprotected display to 16.2% for the readability-priority profile and 5.0% for the high-suppression profile. These results provide evidence of feasibility within the evaluated short-exposure camera link rather than a general guarantee against screen photography.

The evaluation therefore extends beyond favorable single-frame conditions. Fixed image preprocessing, long exposure, 150-frame temporal averaging, digital reconstruction, and VLM probes test how recovery changes as attacker capability increases. These attacks recover substantial information, especially from large-font short snippets, and expose the integration boundary of the approach. A 56-participant study separately measures the human side of the trade-off: readability-priority masking retains high typing accuracy but imposes measurable costs in speed, perceived stability, and visual comfort.

The main contributions are as follows:

1) **System-level physical evidence**: We provide a controlled measurement of short-exposure screen capture using archived physical images, multiple content types and geometries, three OCR engines, explicit sensitive-field recovery, matched profile comparisons, and content-cluster resampling. This evaluation quantifies temporal pixel masking as a system behavior rather than inferring protection from digitally generated subframes alone.
2) **Attack-boundary characterization**: We evaluate fixed preprocessing, long-exposure capture, temporal-average attacks, digital reconstruction, and commercially hosted VLMs. The resulting evidence identifies both the conditions under which conventional OCR recovery decreases and the conditions under which sustained capture or stronger recognition recovers substantial content. Detection and tracking experiments provide secondary cross-task stress tests.
3) **Security–usability trade-off**: We compare readability-priority and high-suppression operating points and conduct a controlled within-subject study with 56 participants. The results connect machine recovery to typing performance, readability, perceived stability, and immediate visual comfort, making the cost of the usability-facing configuration explicit.

The remainder of this paper reviews related work, defines the threat model and system design, and then presents the physical, attack-oriented, and usability evaluations. The final sections discuss the evidence boundaries and conclude the study.

II. RELATED WORK

A. SCREEN VISUAL EAVESDROPPING THREATS

Sensitive screen content can be acquired through direct photography, optical reflections, or non-visual side channels. Backes et al. [11] recovered displayed content from reflections on nearby objects, while Raguram et al. [12] reconstructed typed input from reflections and refractions in a user's surroundings. Electromagnetic attacks remove the need for an optical line of sight: Screen Gleaning recovered security codes from mobile-display emissions [13], and Deep-TEMPEST used learned reconstruction to recover text from unintended HDMI emissions [14]. Camera glasses further increase the likelihood of inconspicuous recording, although recent work by Wang et al. [1] studies wearer–bystander privacy expectations and protection preferences rather than the technical recovery of screen content.

Human-centered studies establish the practical relevance of these threats but do not by themselves characterize machine-readable leakage. An industry-sponsored visual-hacking experiment reported successful information capture in 91% of trials, with the first success occurring within 15 minutes in 49% of trials [2]. Eiband et al. [3] collected 174 accounts of shoulder-surfing incidents and found that observation was commonly opportunistic and often went unnoticed. These studies motivate display privacy, whereas the present work focuses on a narrower technical question: how much static text remains recoverable from short-exposure physical captures, and how that recovery changes under stronger acquisition and post-processing attacks.

B. DISPLAY PRIVACY PROTECTION

Existing defenses intervene at the camera, the display, or the surrounding illumination. ScreenAvoider detects displays and sensitive applications in images so that a cooperating capture or disclosure pipeline can suppress them [15], its protection therefore depends on control of the camera or downstream image handling. Display-side spatial methods instead alter what an observer sees. HideScreen suppresses low-frequency visual cues so that content blends into the background beyond a designed viewing distance [4]. EyeShield re-renders content to preserve close-range readability while reducing recognition at larger distances and viewing angles [16], in addition to user studies, it evaluated recognition on photographs using cloud OCR. Both systems primarily address distance- and angle-dependent shoulder surfing rather than temporally sampled display content. Illumination-based systems such as LiShield [17] and Rainbow [18] exploit rolling-shutter capture by modulating ambient light, but they protect illuminated physical scenes rather than altering the displayed pixels themselves.

Attribution mechanisms address a different stage of the leakage process. General image watermarking embeds identifiers that may survive subsequent processing [19], and screen-shooting-resilient schemes explicitly model the display–camera channel [5]. The mID system similarly creates camera-visible moiré identifiers for tracing leaked screen photographs [20]. These approaches can identify a leakage source after capture, but they are not designed to prevent an attacker from reading the photographed content. Visual cryptography is also conceptually related because it decomposes an image into shares that reveal the secret only when combined [21]. Its original threat model, however, concerns cryptographic share construction rather than time-varying display capture.

Temporal display techniques are the closest predecessors of the present work. Yerazunis and Carbone introduced time-masked privacy-enhanced displays [6]. Wu and Zhai subsequently formalized temporal psychovisual modulation [8], and Gao et al. applied that framework to an information-security display [9]. Lim used persistence of vision to construct browser keypads that resist a single digital screenshot [7]. Kaleido decomposes video into chrominance-complementary frames with controlled luminance variation so that viewers perceive the intended sequence while camera recordings are degraded [10]. Screen-camera communication explores the complementary objective: systems such as spatially adaptive embedding, DeepLight, and passive screen-to-camera communication encode camera-decodable information while limiting perceptual disturbance to human viewers [22]–[24].

These studies establish temporal modulation and multi-frame decomposition as prior mechanisms, the present work does not claim to introduce either concept. The unresolved issue is their security boundary for static sensitive text under modern machine recognition and increasingly capable capture strategies. Existing studies do not jointly measure short-exposure physical capture, multi-engine OCR, vision-language-model recovery, temporal integration, and phase-aware reconstruction. Our implementation also differs from Kaleido in mechanism: it uses randomized discrete pixel partitioning, and its exploratory high-suppression profile deliberately breaks strict per-cycle completeness (§IV-E). Table 1 compares representative systems by protection locus, threat model, and reported evaluation.

C. MACHINE RECOGNITION AND PHYSICAL-WORLD ROBUSTNESS

Research on adversarial examples provides a second point of comparison because it separates human readability from machine recognition. Song and Shmatikov showed that small digital modifications to text images can induce targeted OCR errors while preserving the apparent meaning for human readers [25]. More general digital attacks optimize model-specific perturbations [26]–[28], whereas physical attacks must remain effective after changes in viewpoint, illumination, printing, deformation, and camera processing [29]–[31]. Athalye et al. introduced Expectation over Transformation (EoT) to optimize adversarial examples over a chosen transformation distribution and demonstrated robust physical 3D objects [32]. Surveys of physical attacks and adversarial transferability show that robustness across capture conditions and generalization across models are distinct empirical questions [33], [34].

The display–camera channel adds transformations that are absent from ordinary digital evaluation. Niu et al. showed that simulated moiré patterns from screen photography can act as transferable adversarial perturbations [35]. Conversely, learned demoiréing can recover screen photographs degraded by such artifacts [36], illustrating why incidental optical distortion should not be treated as a stable security guarantee. The base mechanism studied here does not optimize a spatial perturbation against a particular recognizer, it partitions the displayed content over time. Enhanced profiles add noise and stripe or glyph overlays, but their effectiveness is measured after physical capture across multiple OCR and vision-language models rather than inferred from digital attack success. This distinction motivates our system-level evaluation of recovery and attack boundaries.

III. THREAT MODEL

TABLE 1. Scope comparison with representative screen-privacy systems. Entries describe different threat models and evaluation targets, they are not matched-parameter performance claims.

Scheme	Protection locus	Target threat	Mechanism	Reported evidence and boundary
ScreenAvoider [15]	Camera or disclosure pipeline	Incidental capture of sensitive screens	Screen and application recognition, policy-based suppression	Lifelogging-image detection, requires a cooperating capture or disclosure path
HideScreen [4]	Mobile display	Human viewing beyond a designed distance	Spatial suppression of low-frequency cues	Human text/image recognition and usability across viewing distances
Eye-Shield [16]	Mobile display	Human viewing at larger distances and angles	Distance- and angle-aware spatial re-rendering	Human studies and photographed-content recognition, spatial rather than temporal defense
LiShield and Rainbow [17], [18]	Ambient illumination	Rolling-shutter photography of physical scenes	Temporally modulated LED illumination	Camera-image interference, requires controllable illumination and rolling-shutter interaction
Yerazunis and Gao [6], [9]	Displayed image	Incomplete temporal observation	Time masking or temporal psychovisual modulation	Proof-of-concept security displays, no modern OCR/VLM attack suite
Kaleido [10]	Displayed video	Continuous mobile-camera recording	Chrominance-complementary frames and luminance variation	PSNR, structural similarity, and subjective quality for protected video
This work	Static screen text	Short-exposure capture followed by machine recognition	Randomized pixel partitioning, high-suppression profile breaks completeness	OCR/VLM recovery, temporal integration, phase-aware reconstruction, and usability

A. THREAT SCENARIO AND PROTECTED ASSETS

The attacker is assumed to photograph the display from a direct or oblique angle and process the image, without compromising the host computer, application, or source document. Protected assets consist of confidential text-based information. The authorized user is expected to read and interact with the protected display directly. Our system does not prevent image acquisition, its narrower objective is to reduce machine-recoverable text in selected physical capture conditions while retaining usable visual content for the authorized user.

B. ATTACKER CAPABILITIES AND KNOWLEDGE

We organize the evaluated attacker along acquisition, temporal coverage, and recognition dimensions rather than as a strictly monotonic hierarchy. The *target attacker* obtains a short-exposure snapshot and applies conventional per-image OCR, this tier defines the primary feasibility claim. An *enhanced attacker* can vary viewing geometry, apply deterministic image preprocessing, select among recognition engines, or use a vision-language model (VLM). A *boundary attacker* can extend the exposure window, record a sustained sequence, align and aggregate multiple frames, or perform phase-aware reconstruction. The fixed preprocessing oracle covers the full short-exposure capture pool (§V-A), whereas the VLM probes are restricted to the high-suppression profile at on-axis geometries (§V-D). The shared capture, geometry, and preprocessing settings used for these tiers are specified in §V-A. The attack-specific configurations appear with the corresponding experiments, in particular the digital reconstruction attacks in §V-G2. The corresponding attack results are reported in §V-B–§V-G2.

We adopt a public-algorithm threat model: the attacker knows the temporal masking algorithm, subframe count n, refresh rate, profile composition, and candidate period parameters. Pixel-to-slot assignments and output permutations are randomized per cycle by a cryptographically secure pseudorandom number generator (CSPRNG; §IV). The attacker is not assumed to receive the current key or exact starting phase before capture, but neither value is treated as a security anchor. Temporal averaging can accumulate information without either value, and an adaptive attacker can estimate alignment from a captured sequence. Once registered observations cover a complete complementary cycle, key secrecy cannot prevent reconstruction in the ideal linear model.

C. SECURITY OBJECTIVES AND EXPLICIT BOUNDARIES

The primary confidentiality outcomes are OCR character recovery, whole-text exact match, and exact recovery of preidentified sensitive fields. The corresponding utility objective is direct human readability and interaction performance, assessed separately through typing behavior and subjective ratings. Under short exposure, the high-suppression profile uses descriptive targets of conventional-OCR character recovery $< 10\%$ and exact match $= 0\%$, the readability-priority profile targets a relative character-recovery reduction of at least 80% and exact match $< 1\%$. These thresholds organize the empirical comparison and are not formal security guarantees or device-independent deployment criteria. The high-suppression profile is an exploratory security-oriented stress point, whereas the user study evaluates the readability-priority profile.

The claimed protection scope excludes digital screenshots, host compromise, and source-document leakage. It also does not promise resistance to full-cycle exposure, registered multi-frame aggregation, phase-aware reconstruction, unconstrained learned restoration, or VLM-based inference. These stronger attacks are evaluated to identify where recovery increases or the protection fails, which is central to the attack-boundary contribution. Generalization beyond the tested display–camera link is not assumed. Object detection and tracking are reported only as secondary cross-task stress tests, while ΔE_{00} and the structural similarity index (SSIM) describe digital reconstruction quality rather than confidentiality.

IV. SYSTEM DESIGN

A. TEMPORAL SUBFRAME DECOMPOSITION

Given a normalized original frame $\mathbf{I} \in [0,1]^{H\times W\times C}$, we generate binary masks $\mathbf{M}_1,\ldots,\mathbf{M}_n$ satisfying $\sum_{k=1}^{n}\mathbf{M}_k = \mathbf{1}$, and set $\mathbf{I}_k = \mathbf{I}\odot\mathbf{M}_k$. This yields:

- **Digital-domain completeness**: $\sum_{k=1}^{n}\mathbf{I}_k = \mathbf{I}$;
- **Mutual exclusivity**: each source-image pixel is assigned to exactly one of the n binary masks. Accordingly, its unmodified image contribution appears in exactly one base subframe.

The distinction between ‘‘summation’’ and ‘‘temporal averaging’’ matters: on an ideal linear display with equal slot durations and equal peak luminance, the average radiance over one cycle is $\mathbf{I}/n$, not $\mathbf{I}$, so digital-domain completeness means only that subframes can be re-summed, not that real display luminance has been losslessly preserved.

Pixel assignments are generated from a ChaCha20 [37] CSPRNG, with an HMAC-SHA256-derived subkey per cycle, rejection sampling mapping the byte stream to unbiased slot indices, and a separate Fisher–Yates shuffle [38] randomizing subframe playback order. A chi-square slot-count check ($\alpha = 0.01$, df $= n-1$, 3 at the evaluated $n = 4$), used only as an implementation sanity check with at most 5 resampling attempts, is not evidence of cryptographic strength. Per-cycle randomization avoids a fixed, periodically repeated pixel-to-slot pattern across cycles, but it cannot prevent linear reconstruction of a registered full cycle. This ChaCha20-based generator drives the Python physical-capture pipeline. The user-study web implementation reproduces the same mask structure, rejection sampling, and playback shuffling with a deterministic seeded 32-bit generator so that stimuli are exactly repeatable in the browser. And it has no CSPRNG properties (§V-E). Fig. 2 illustrates the complete pipeline.

B. COMPLEMENTARY ADVERSARIAL NOISE

We overlay adversarial noise $\mathbf{N}_1,\ldots,\mathbf{N}_n$ on the subframes, satisfying:

$$\sum_{k=1}^{n}\mathbf{N}_k = \mathbf{0} \tag{1}$$

Equation (1) holds strictly only in the ideal linear domain prior to clipping and the display transfer function. Motivated by prior work showing that small text-image perturbations can significantly mislead OCR systems [25], each cycle uses an available stored perturbation template when available, and otherwise a proxy-gradient perturbation generated by single-step fast gradient sign method (FGSM) [26] or iterative projected gradient descent (PGD) [27] with sign updates under an $\varepsilon = 8/255$ bound. The configured proxy set rotates across eight OCR and detector targets. Depending on availability, gradients come from the EasyOCR recognition network, a differentiable shadow objective, or a deterministic image-space surrogate. The resulting noise is then decomposed complementarily in the temporal domain. This mechanism is a practical transfer-oriented defense component, not a claim of white-box optimality for every recognizer or detector. The user-study web implementation substitutes a deterministic image-space texture of comparable amplitude for these model-gradient perturbations (§V-E). Because displays cannot emit negative radiance, the implementation uses a pedestal level and numerical clipping. Consequently, the noise is not guaranteed to cancel physically.

The components serve distinct functions in this system. The random dot-pattern mask is the primary temporal mechanism, while complementary noise is an optional ideal-domain component. Meanwhile, stripe/glyph overlays together with partial inversion are nonlinear additions used by the enhanced profiles below. The archived composite-profile definitions preserve correspondence with the captured data, and the physical analysis compares both individual ablations and complete profiles.

C. LUMINANCE MODEL ACROSS RENDERING PATHS

Because each pixel is illuminated only once per n-slot cycle, restoring the original cycle-averaged luminance in a linear-light model would require a factor-n light-output increase during the active slot. No evaluated path implements per-slot backlight control. The physical-capture playback fixes the composition gain at $\gamma = 1$ under fixed panel settings, leaving the linear-model cycle average at $\mathbf{I}/n$. The user-study web implementation instead applies a pixel-space gain of $1.1n$ with clipping to $[0, 255]$, which partially compensates the $1/n$ duty cycle but clips bright content. Its stimuli are therefore not photometrically identical to the physical-capture condition. Digitally integrated reconstructions are normalized before comparison, and CIEDE2000 ΔE_{00} [39] and SSIM [40] quantify these digital reconstructions only. None of these settings establishes radiometric equivalence on a real panel.

D. TIMING AND BANDWIDTH

For a basic cycle containing only n complementary subframes, taking 60 Hz as the minimum cycle rate requires $f_r \geq 60n$, at $n = 4$ this corresponds to 240 Hz. Including one inversion frame changes the requirement to $f_r \geq 60(n+1)$, dropping the actual cycle rate to only 48 Hz at 240 Hz. Thus, while 240 Hz meets the basic four-subframe configuration, it falls short of the inversion-frame configuration's 60 Hz cycle target. Because human flicker perception also depends on luminance, contrast, retinal eccentricity, spatial content, and eye movements [41], [42], nominal cycle rate alone cannot predict comfort.

Whether an observer integrates complementary subframes into a perceptually complete image depends on the temporal contrast sensitivity function (TCSF) rather than on a single critical flicker fusion (CFF) cutoff. With high-frequency spatial edges, flicker can remain visible above 500 Hz [42]. The

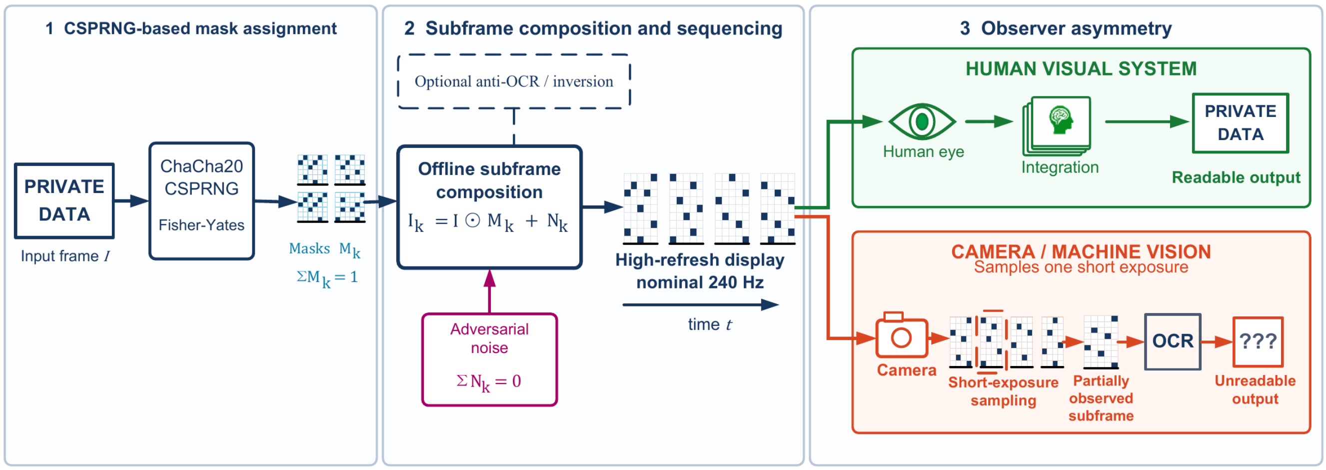

FIGURE 2. Temporal pixel masking system pipeline from source content to protected display. The input frame is decomposed into CSPRNG-assigned complementary pixel masks. Optional complementary OCR-oriented noise, profile-dependent stripe/glyph overlays, and an optional partial inversion frame are then inserted before high-refresh playback. A human observer is expected to integrate multiple slots into a readable view, whereas a short-exposure camera may capture only a sparse subframe. Long exposure and video temporal averaging can accumulate multiple slots and are evaluated separately.

48 Hz full-cycle rate of the inversion configuration may therefore produce visible flicker for some observers, the 60 Hz basic configuration cannot be assumed flicker-free, and the user study in §V-E evaluates readability and immediate comfort under these configurations.

E. OPERATING PROFILES AND CONFIGURABILITY

Our work is configurable, and the evaluation uses named, composable operating profiles rather than treating one parameter setting as universally appropriate. Unless otherwise stated, the protected profiles use $n = 4$ temporal frames, with complementary perturbation included as indicated in Table 2. The table defines the components and intended evaluation role of each profile, including the unprotected baseline. The exact parameter instances used in the physical-capture evaluation are reported separately in Section V-A. This separation distinguishes a reusable system interface from the particular configurations evaluated in this paper.

The nonlinear profile combinations are intentionally not interpreted as the sum of their isolated components. Compared to the readability-priority profile, the high-suppression profile uses coarser 2×2 pixel masks and denser stripes, significantly increasing stripe and glyph strengths. It also adds a dense line mesh and false strokes to aggressively degrade text recovery at the expense of readability. Because its visible artifacts may cause visual fatigue, this profile is unsuitable for default deployment and is included solely for experimental evaluation.

F. PARTIAL INVERSION FRAME

A long-exposure attack may cover a complete display cycle. To perturb, rather than eliminate, full-cycle accumulation, the evaluated offline-playback and web pipelines append a **partial inversion frame** $\alpha(\mathbf{1} - \mathbf{I})$ (normalized domain $\mathbf{I} \in [0, 1]$, 8-bit form $\alpha(255 - \mathbf{I}_8)$) after each n-subframe cycle, implemented as amplitude scaling over one additional display slot because vsync playback holds slot duration fixed. Higher α changes both the integrated signal and the digital reconstruction's luminance and contrast.

A real-capture ablation across $\alpha \in \{0, 0.2, 0.3, 0.5, 1.0\}$ (§V-C) informs this choice, the readability-priority setting of $\alpha = 0.2$ is a modeled design decision, not a validated optimum.

G. IMPLEMENTATION AND REPRODUCIBILITY

The proof of concept (PoC) operates at the application layer in Python (with NumPy, PyCryptodome, pygame/SDL, mss, ModernGL, etc.) and separates two rendering paths. One is the evaluated physical-capture path. It pre-generates all subframes offline with a CPU compositor and presents the sequence through pygame, requesting VSync and applying a software frame-rate cap when VSync is unavailable, no per-slot synthesis occurs during display. The other path is for user study. The user study runs a third, independent JavaScript/Canvas2D implementation (§V-E). Mask keys default to 32 bytes from the operating-system CSPRNG, and a fixed key can be supplied for regeneration. The playback metadata records permutations, the per-cycle noise schedule, and profile parameters, but not the key, so reproduction relies on the archived stimuli and explicit configurations rather than on command-level replay. No operating-system, display-driver, or backlight-level integration is implemented.

V. EXPERIMENTAL EVALUATION

A. EXPERIMENTAL SETUP

Hardware platform: Most experiments used a desktop workstation with an Intel Core i7-8700K (3.70 GHz), 32 GB RAM, and an NVIDIA TITAN Xp (12 GB VRAM). The display was

TABLE 2. Component-Level Definitions of the Evaluated Operating Profiles. Exact Parameter Instances Are Reported in Section V-A.

Profile	Temporal mask	Complementary noise	Additional display artifacts	Intended evaluation role
Original (unprotected)	–	–	–	Unprotected baseline
Mask only	✓	–	–	Isolate temporal decomposition
Mask + noise	✓	✓	–	Isolate complementary perturbation
Strong (anti-OCR)	✓	✓	Stripe and glyph overlays	Isolate the effects of stripe and glyph overlays
Readability-priority	✓	✓	Stripe and glyph overlays with partial inversion	Evaluate the complete configuration for usability
High-suppression	✓	✓	Enhanced overlays and partial inversion	Exploratory recovery-boundary stress condition

an AOC 24G51Z driven at a nominal 240 Hz. Each nominal 240-Hz slot lasts 4.17 ms, while the basic and inversion-augmented cycles are 60 and 48 Hz.

Capture setup: All physical captures used one eMeet SmartCam S600 USB webcam with its standard UVC driver. Camera frames underwent perspective correction and content cropping before recognition. Per-capture metadata store exposure labels derived from DirectShow/UVC $\log_2$ controls.

The camera was placed at 0.5 m, 1 m, and 1.5 m from the display, rotated to $0°$, $15°$, and $30°$ at each distance, providing nine distance–angle configurations for the physical evaluation. The manufacturer specifies an 8-megapixel 4K sensor and 1080p/60 fps output [43].

All nine geometries used the same control values: -8 for short-exposure stills and video frames (nominally 3.91 ms), and -5 for long-exposure stills (31.25 ms). These values are control-derived labels. The display-brightness setting remained fixed, profiles were acquired in a fixed order, and video was captured nominally at 60 fps. Auto-exposure, gain, white balance, focus, and panel luminance were neither locked nor independently measured. Accordingly, the results characterize the evaluated display–camera link rather than an isolated temporal mechanism. The primary OCR inputs are geometrically corrected content crops.

Test content: Each position used 12 fixed items: two account strings, two code snippets, two digit strings, one English sentence, two mixed-language snippets, two English paragraphs, and one CET6 (College English Test Band 6) document. The CET6 page and two mixed-language items include Chinese characters. Three repeats give 36 captures per profile–attack–position cell and 324 across 9 positions before disclosed repeats. Detection and tracking use separate subsets as cross-task stress tests.

Evaluated configurations: Table 3 reports the exact profile instances used in the physical-capture evaluation. These settings are experimental configurations rather than universal implementation defaults. In particular, the readability-priority capture condition invokes the `strong` profile with explicit stripe and glyph opacity overrides of 0.10 and 0.12, respectively, and a partial inversion frame with $\alpha = 0.20$; the reusable `strong` command-line default uses different overlay amplitudes. Relative to readability-priority, high-suppression further strengthens the masking configuration by increasing the cell size from 1 to 2 pixels, narrowing the stripe width from 10 to 6 pixels, and raising the stripe and glyph opacities from 0.10 and 0.12 to 0.42 and 0.55, respectively, while retaining the same partial inversion strength of $\alpha = 0.20$.

Evaluation metrics: OCR character recovery is defined as $\max(0, 1 - \text{CER})$, where CER is character error rate. Exact match uses case-insensitive full-text comparison after whitespace removal. Explicit-field recovery uses 18 fields fixed before scoring across 9 field-bearing content items. A field counts as recovered only when its complete Unicode-normalized form appears in at least one engine output for that capture; case, whitespace, and punctuation are ignored. We report field-level micro recovery and sample-level macro recovery separately. Leak rate is the proportion of samples with character recovery $\geq$ 20% and is only a descriptive threshold.

OCR aggregation and uncertainty: Character and whole-text metrics are maximized independently across the three engines for each capture. Explicit fields use the union of fields recovered by the three engines. Duplicate captures with the same profile, content, position, and repeat are averaged before matching. Thus, the 324 units in each primary profile comprise 12 content items, 9 geometries, and 3 repeats; they are not treated as 324 independent population samples. The primary comparison uses keys present in all three profiles. Paired contrasts resample the 12 content items as clusters with 2,000 resamples and seed 20260612, holding the tested geometries fixed. These intervals quantify sensitivity across the archived content clusters.

Common-setting analysis: All 9 geometries define the primary fixed-setting result (Table 4). After duplicate averaging and three-profile matching, each profile contains $N = 324$ units. Table 6 separately reports the unbalanced all-available capture summary before duplicate averaging and profile matching.

Fixed preprocessing stress test: We evaluated Tesseract, EasyOCR, and Surya on all 1,107 primary short-exposure images spanning the 9 geometries. The fixed input grid comprised raw, gamma 0.5, luminance contrast-limited adaptive histogram equalization (CLAHE; 1st–99th percentile stretch, clip limit 4.0, 8×8 tiles), unsharp masking ($\sigma = 1.0$, weights $1.7/{-0.7}$), Gaussian adaptive thresholding (block 31, constant 5), and $2\times$ bicubic upscaling. Parameters were fixed before scoring. For each metric and capture, the attacker retained the best engine–input pair; duplicate captures were then averaged before three-profile matching. The complete matrix contains 19,926 cells (1,107 images $\times$ 6 input forms $\times$ 3 engines) with no OCR errors. Nonraw cells used Tesseract

TABLE 3. Exact Profile Instances Used in the Physical-Capture Evaluation ($n = 4$ for All Masked Profiles). The Readability-Priority Row Is the Archived `deployed` Condition and Explicitly Overrides the Reusable `strong` Defaults.

Profile	Mask	Noise	Cell (px)	Stripe width (px)	Stripe α_s	Glyph α_g	Inv. α
Original (unprotected)	–	–	–	–	–	–	–
Mask only	✓	–	1	–	–	–	–
Mask + noise	✓	✓	1	–	–	–	–
Strong (anti-OCR)	✓	✓	1	10	0.10	0.12	–
Readability-priority	✓	✓	1	10	0.10	0.12	0.20
High-suppression	✓	✓	2	6	0.42	0.55	0.20

5.4.1 through `pytesseract 0.3.13`, EasyOCR 1.7.2, and Surya 0.14.7; transformed cells were evaluated on CUDA or CPU with the same software versions. The resulting oracle measures a reproducible fixed-grid preprocessing attack on the nine-geometry primary pool.

Digital reconstruction proxies: ΔE_{00} is the mean CIEDE2000 color difference between the original image and the full-cycle digitally reconstructed image; SSIM [40] is their mean structural similarity. Both metrics support within-paper comparison of digitally reconstructed profiles.

B. REAL-CAPTURE OCR EXPERIMENT

The physical archive comprises 10,575 images (1,175 per position), including ablations, parameter sweeps, and main attack conditions. Readability-priority includes a second capture round for 5 of 12 items, giving 459/153 all-position short/video images rather than 324/108. Duplicate keys are averaged before matched contrasts. Surya [44], EasyOCR [45], and Tesseract [46] produced 31,725 engine-level records. The Surya rerun used `surya-ocr 0.14.7`; the primary pipeline also used Tesseract 5.4.1 (`pytesseract 0.3.13`) and EasyOCR 1.7.2. Each geometry uses an archived four-point homography and output canvas, followed by the recorded content crop. The primary OCR pipeline adds no adaptive enhancement. Profiles were collected in the order original, mask-only, mask+noise, Strong, readability-priority, and high-suppression. The main comparison uses matched units from all 9 geometries, while Table 6 retains all available captures and therefore remains unbalanced across profiles. Fig. 3 shows digital reconstructions and matched captures at 1.0 m/0°.

Key findings:

1) **Primary fixed-setting evidence**: Table 4 is the main short-exposure estimate. On the same 324 matched units per profile (12 content items × 9 geometries × 3 repeats), mean recovery is 94.1% unprotected, 16.2% for readability-priority, and 5.0% for high-suppression. The paired content-cluster contrasts are 78.0 percentage points ([75.5, 80.3]) and 89.1 points ([85.0, 92.4]). Table 6 reports the all-available capture summary, in which the readability-priority short-exposure mean is 15.1% because disclosed repeat captures remain separately weighted before duplicate averaging.
2) **Fixed preprocessing defines a stronger OCR attack**: Across the same 324 matched units per profile, the raw three-engine oracle yields 94.1% unprotected, 16.2% readability-priority, and 5.0% high-suppression recovery. Allowing the five fixed transforms raises these values to 95.5%, 37.2%, and 13.2%, respectively (Table 8). The paired unprotected-minus-readability contrast is 58.4 percentage points ([53.4, 62.8]), while the unprotected-minus-high-suppression contrast is 82.4 points ([78.1, 85.9]). The complete 19,926-cell matrix has no OCR errors.
3) **Profiles show distinct component and target behavior**: The mask-only condition reaches 19.2% in the all-available archive, while mask+noise reaches 25.6% and Strong reaches 20.7%, revealing nonmonotonic component effects. Across the nine-geometry matched pool, readability-priority retains 15.7% field-micro exact recovery (Table 5). Its P50, P75, P90, P95, and P99 character-recovery values are 9.2%, 18.7%, 43.5%, 61.4%, and 94.8%, respectively; the corresponding high-suppression values are 1.6%, 8.3%, 14.1%, 18.7%, and 22.2%. High-suppression reaches a 5.0% matched mean at the common setting, meeting both its descriptive $< 10\%$ character target and the exact-match $= 0\%$ target: no short-exposure capture produced an exact match (Table 6). For readability-priority, the raw common-setting reduction is $(94.1 - 16.2)/94.1 = 82.8\%$; the all-available exact-match rate is 0.4%, meeting the $\geq 80\%$ relative reduction and $< 1\%$ exact-match targets. Under the fixed preprocessing grid, the relative reduction is $(95.5 - 37.2)/95.5 = 61.0\%$, falling short of the same threshold.
4) **Integration attacks recover substantial text** (Fig. 4): Across the nine geometries, the readability-priority profile reaches 60.9% char / 36.4% exact under nominal long exposure and 71.1% char / 42.5% exact under the 150-frame mean. The latter aggregates 150 frames for each content–position video unit, corresponding nominally to 2.5 s at 60 fps. Weak inversion ($\alpha = 0.2$) produces little integration reduction, and mask-only and Strong are similar under the 150-frame mean (79.0% versus 78.6%). High-suppression shows lower integration recovery, although substantial text remains recoverable.
5) **Long exposure produces a position-dependent reversal**: Under the common nominal 31.25 ms setting, the unprotected long-exposure baseline is 47.3%

TABLE 4. Primary Matched Short-Exposure OCR Comparison at the Common Nominal 3.91-ms UVC Setting. Duplicate Captures Are Averaged Before Matching; $N = 324$ per Profile Comprises 12 Content Items $\times$ 9 Geometries $\times$ 3 Repeats. Paired contrasts resample the 12 content items as clusters while holding the 9 tested geometries fixed; they do not treat the 324 units as independent population samples.

Profile	N	Mean char recovery	Unprotected − profile	Interpretation
Original (unprotected)	324	94.1%	–	Common-setting baseline
Readability-priority	324	16.2%	78.0 pp [75.5, 80.3]	Conventional-OCR reduction, with upper-tail failures
High-suppression	324	5.0%	89.1 pp [85.0, 92.4]	Exploratory stress profile; meets descriptive $<$ 10% target

TABLE 5. Explicit Sensitive-Field Exact Recovery Across the Nine-Geometry Matched Pool. The 18 Fields Were Fixed Before Scoring; 243 of 324 Units per Profile Contain at Least One Field.

Profile	Field micro (486 opp.)	Sample macro
Original (unprotected)	90.1%	85.5%
Readability-priority	15.7%	15.6%
High-suppression	1.0%	1.0%

char, below the readability-priority profile's 60.9%. Geometry-specific values are nonmonotonic, and visible clipping occurs in some full-brightness captures. Because the camera's automatic state was not independently measured, residual stripe/glyph edges and geometry-specific saturation or contrast compression are candidate explanations rather than an isolated causal account. Short-exposure recovery remains lower for readability-priority at every position.

6) **High-suppression reduces integration leakage**: This exploratory stress profile yields 9.3% char [7.9, 10.9] / 0% exact for nominal long exposure and 47.9% char [41.8, 54.0] / 5.6% exact [1.9, 10.2] under the 150-frame mean. Brackets are capture-resampling summaries, and the pooled mean includes considerable inter-position variation. The user study evaluates readability-priority rather than this visibly stronger profile.
7) **Geometric trends**: Across 9 conditions (0.5–1.5 m $\times$ 0–30°), protected short-exposure recovery remains below unprotected recovery. Geometry-specific values are reported descriptively because the archived design is unbalanced. A round-selection sensitivity analysis over all 12 content-item strata gives readability-priority short-exposure recovery of 16.6% when round 1 is selected for the five repeated items and 15.8% when round 2 is selected; across the five repeated items alone, the corresponding means are 13.5% and 11.7%. The profile-level pattern remains unchanged.

Per-engine results: Table 7 breaks down pooled raw-input short-exposure results by engine. The engines differ widely in unprotected recovery (Surya 37.1%, EasyOCR 79.5%, Tesseract 84.3%). Under Strong, recovery ranges from 2.1–16.3%; under readability-priority, 3.2–11.7%; under high-suppression, 1.8–2.0%. Table 8 and Fig. 5 use the full nine-geometry matched pool and show that every engine benefits from at least one fixed transform. For readability-priority, recovery rises from 3.2% to 10.4% for Tesseract, 12.6% to

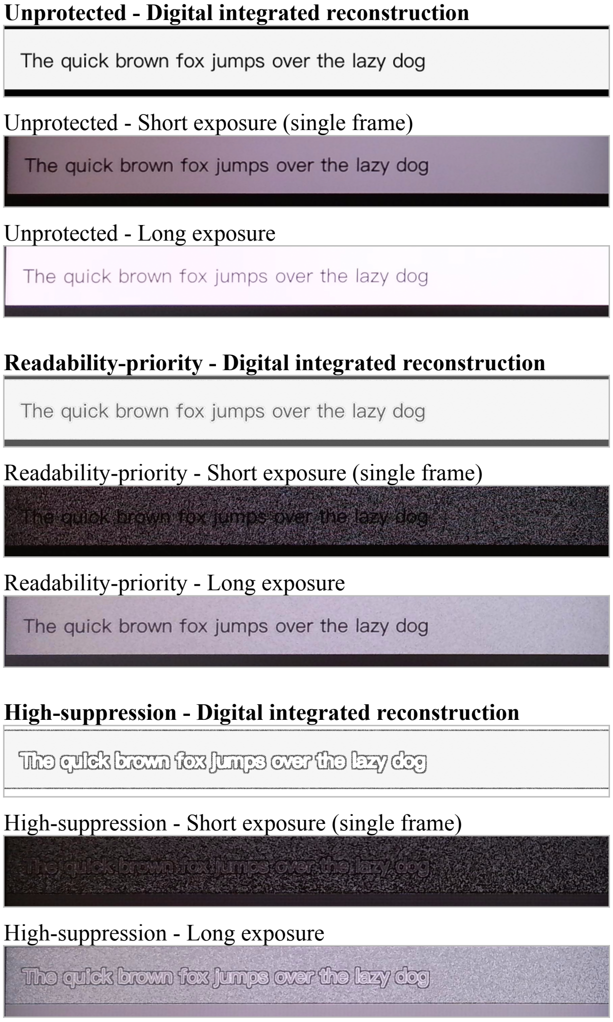

FIGURE 3. Real-capture montage at 1.0 m and 0°. Within each vertically stacked profile group (unprotected, readability-priority, and high-suppression), panels show, from top to bottom, the digital integrated reconstruction, 3.91 ms short-exposure capture, and 31.25 ms long-exposure capture. Camera panels are identically cropped from the archived raw camera JPEGs with no geometric correction or brightness gain, so they differ from the geometrically corrected crops used for OCR evaluation. Digital panels use the same 1920 $\times$ 1080 black playback canvas and brightness-aligned full-cycle integration for each profile (for the unprotected profile this integration equals the displayed source frame); the readability-priority and high-suppression reconstructions include the recorded $\alpha = 0.2$ inversion slot. These digital views are not direct evidence of human readability or visual comfort.

31.0% for EasyOCR, and 3.6% to 18.3% for Surya. For high-suppression, the corresponding changes are 1.9% to 7.6%,

TABLE 6. Single-UVC Real-Capture OCR All-Available Summary (Best-of-Engine; N = Captures per Row; All 9 Geometries). This archive is unbalanced because the readability-priority profile includes disclosed repeat captures. Brackets are 95% capture-resampling summaries, not population confidence intervals. Each video row averages 150 frames for one content–position video unit (nominally 2.5 s at 60 fps). The duplicate-averaged matched estimate is Table 4.

Profile	Attack mode	N	Char recovery [95% summary]	Exact match	Leak rate
Original (unprotected)	Short exposure	324	94.1 [93.1, 95.0]%	65.7%	99.4%
	Video (150-frame mean)	108	79.9 [73.5, 86.0]%	61.1%	86.1%
	Long exposure	324	47.3 [42.8, 51.9]%	18.8%	58.6%
Mask only	Short exposure	324	19.2 [17.0, 21.4]%	0.3%	30.6%
	Video (150-frame mean)	108	79.0 [73.0, 84.9]%	48.1%	88.0%
	Long exposure	324	67.2 [63.0, 71.5]%	35.5%	75.0%
Mask + noise	Short exposure	324	25.6 [22.5, 28.8]%	2.8%	36.1%
	Video (150-frame mean)	108	79.2 [73.3, 85.0]%	49.1%	88.9%
	Long exposure	324	68.0 [63.7, 72.4]%	39.8%	76.2%
Strong (anti-OCR)	Short exposure	324	20.7 [18.2, 23.3]%	0.6%	32.4%
	Video (150-frame mean)	108	78.6 [72.5, 84.5]%	50.9%	88.9%
	Long exposure	324	61.6 [57.1, 66.1]%	39.8%	71.3%
Readability-priority	Short exposure	459	15.1 [13.3, 17.1]%	0.4%	20.5%
	Video (150-frame mean)	153	71.1 [65.7, 76.2]%	42.5%	85.0%
	Long exposure	324	60.9 [56.6, 65.6]%	36.4%	69.1%
High-suppression	Short exposure	324	5.0 [4.3, 5.8]%	0.0%	3.7%
	Video (150-frame mean)	108	47.9 [41.8, 54.0]%	5.6%	76.9%
	Long exposure	324	9.3 [7.9, 10.9]%	0.0%	14.2%

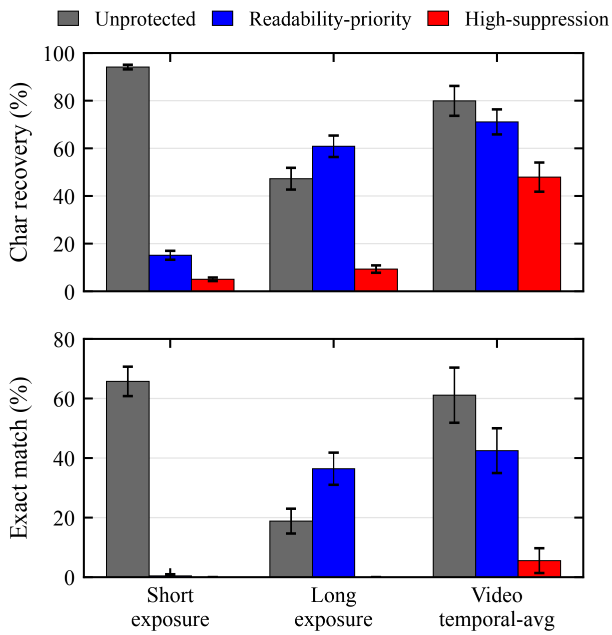

FIGURE 4. Best-of-engine OCR character recovery and exact match for three representative real-capture profiles (unprotected, readability-priority, and high-suppression) under short exposure, long exposure, and a 150-frame video temporal mean, pooled across the 9 tested distance–angle configurations. This all-available summary differs from the matched comparison in Table 4 because readability-priority sample counts include the disclosed five-item repeat. Lower bars indicate less recovered text. The long-exposure and video bars expose the integration boundary: the readability-priority profile retains substantial recoverable content even when its short-exposure bar is low.

2.0% to 5.8%, and 1.8% to 6.1%.

TABLE 7. Real-Capture Short-Exposure OCR: Per-Engine Results (Pooled Across 9 Geometric Conditions)

Profile	Engine	Char (%)	Exact (%)	N
Original	Surya	37.1	22.8	324
	EasyOCR	79.5	27.8	324
	Tesseract	84.3	47.8	324
Mask only	Surya	4.2	0.3	324
	EasyOCR	15.6	0.0	324
	Tesseract	3.2	0.0	324
Mask + noise	Surya	11.0	2.8	324
	EasyOCR	18.1	0.0	324
	Tesseract	4.8	0.0	324
Strong (anti-OCR)	Surya	6.1	0.6	324
	EasyOCR	16.3	0.0	324
	Tesseract	2.1	0.0	324
Readability-priority	Surya	3.2	0.4	459
	EasyOCR	11.7	0.0	459
	Tesseract	3.3	0.0	459
High-suppression	Surya	1.8	0.0	324
	EasyOCR	2.0	0.0	324
	Tesseract	1.9	0.0	324

TABLE 8. Character Recovery Across the Nine-Geometry Matched Pool Under the Fixed Preprocessing Grid. Each Cell Is Raw Input → Per-Capture Grid Oracle (%); N = 324 per Profile After Duplicate Averaging and Matching.

Engine	Original	Readability	High-supp.
Tesseract	84.3 → 91.4	3.2 → 10.4	1.9 → 7.6
EasyOCR	79.5 → 87.4	12.6 → 31.0	2.0 → 5.8
Surya	37.1 → 54.0	3.6 → 18.3	1.8 → 6.1
Best of three	94.1 → 95.5	16.2 → 37.2	5.0 → 13.2

C. INVERSION-STRENGTH ABLATION

A real-capture inversion-strength ablation (Table 9; pipeline per §V-B) found 68.6% recovery at $\alpha = 0$ and 67.0%

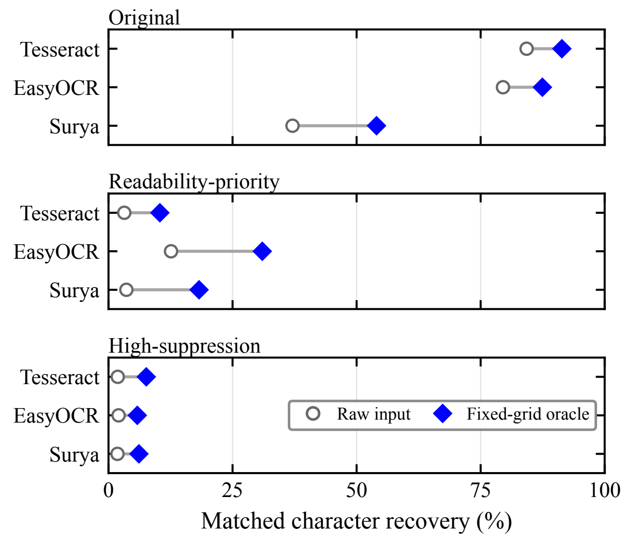

FIGURE 5. Raw-input versus fixed-grid-oracle character recovery for each OCR engine across the nine-geometry matched pool ($N = 324$ per profile). The oracle selects the best of raw plus five predeclared transforms separately for each capture and metric. Lines show within-engine increases, not uncertainty intervals. Exact values appear in Table 8.

TABLE 9. Inversion Frame Strength α Ablation on the Fixed Strong Overlay ($\alpha_s = 0.10$, $\alpha_g = 0.12$): Real-Capture Best-of-Engine Character Recovery (Mean and 95% Capture-Resampling Interval, Five Account/Code/CET6 Items Pooled Across 9 Geometric Conditions; $N = 135$ per Setting for Long Exposure, $N = 45$ for Video Temporal Averaging). This Five-Item Subset Differs from the 12-Item Readability-Priority Rows in Table 6.

α	Long-exp. char (%)	Video avg. char (%)
0.0	68.6 [63.1, 73.9]	72.7 [63.3, 81.1]
0.2	67.0 [61.3, 72.7]	69.8 [59.9, 78.7]
0.3	72.0 [66.2, 76.9]	64.3 [54.0, 73.6]
0.5	61.7 [55.6, 67.7]	66.4 [56.5, 75.2]
1.0	18.1 [14.0, 22.0]	28.4 [20.5, 36.9]

at $\alpha = 0.2$. Point estimates across $\alpha \in [0, 0.5]$ ranged from 61.7% to 72.0%; the capture-resampling intervals are descriptive and interval overlap is not an equivalence test. The ablation holds the Strong overlay fixed and uses a five-item account/code/CET6 subset. Its $\alpha = 0.2$ value ($N = 135$) is not a repeated estimate of the readability-priority row. Near-full inversion ($\alpha = 1.0$) reduced observed recovery to 18.1% while severely degrading the digital integrated view. Under the 150-frame temporal mean, settings at $\alpha \leq 0.5$ retained 64–73% recovery. The readability-priority choice of $\alpha = 0.2$ is a modeled design decision, not a validated optimum.

D. REAL-CAPTURE VLM PROBES

To probe the attack boundary against stronger recognizers, we applied three commercially hosted VLM endpoints exposed through the SiliconFlow application programming interface (API): Qwen3-VL-32B-Instruct [47], Kimi-K2.6 [48], and GLM-4.5V [49]. These probes characterize recovery on the evaluated photographs rather than serving as a cross-platform VLM benchmark.

The probes cover 0° on-axis at three distances. The nominal 60 fps video was split into 12 equal segments and aggregated four ways: (1) *single best*: the single frame within each segment with the highest grayscale contrast; (2) *150-frame mean*: the pixel-wise average of the 150 frames assigned to that segment (nominally 2.5 s); (3) *window-mean best*: a sliding 30-frame mean, keeping the window with the highest mean luminance; and (4) *max projection*: the per-pixel temporal maximum across all frames in the segment. The 0.5 m run includes 156 inputs: 36 unprotected short-exposure baselines, 36 high-suppression short- and long-exposure inputs each, and these four aggregate views (12 segments each). The 1.0 m and 1.5 m runs each include 120 high-suppression inputs: 36 short- and long-exposure inputs each plus four video aggregate types (12 segments each). VLM inputs and the OCR best-of-engine (BoE) reference in Table 10 use the same file batch with nominal 3.91/31.25 ms labels. All VLM calls use raw camera JPEGs.

We use the 1.5 m run as primary cross-model evidence; the 1.0 m run enters the short-exposure distance sweep as supplementary data. At 1.0 m, Qwen and Kimi long-exposure exact match are 84.4% and 74.3%, and the three models yield 70.0–83.3% exact match under the 150-frame video mean. GLM long-exposure exact match is 3.3% at 1.0 m versus 58.3% at 1.5 m. For compactness, Table 10 reports only the 150-frame video mean at 0.5 m; the 1.5 m run reports all four aggregate views to show view-selection sensitivity.

All calls used SiliconFlow's OpenAI-compatible API. The recorded provider identifiers are `Qwen/Qwen3-VL-32B-Instruct`, `Pro/moonshotai/Kimi-K2.6`, and `zai-org/GLM-4.5V`. Qwen3-VL and GLM-4.5V have public technical reports [47], [49]. For Kimi-K2.6, the public model card [48], the provider endpoint identifier, and archived request metadata form the reproducibility record. Decoding temperature is 0 and maximum output is 4,096 tokens. The shared prompt requests only visible text and prohibits guessing; ground truth is not provided. Full prompts, parameters, and per-call outputs are archived. Metrics follow §V-A and are pooled at the capture level.

Table 10 presents the pooled VLM probe results, where short-exposure rows show distance variation while long-exposure and video aggregate rows expose integration attack boundaries; Table 11 breaks down 0.5 m high-suppression short-exposure character recovery by content type.

Key findings:

1) **VLMs recover substantially more text than conventional OCR**: In the 0.5 m probe, best-of-engine OCR yields 8.4% char / 0% exact on high-suppression short exposure, while Qwen3-VL achieves 91.5% char and exactly recovers 28/36 captures. At 1.5 m, the three models retain 33.3–47.2% exact / 61.5–70.0% char. The gap between conventional OCR and modern VLMs is large.
2) **VLM recovery concentrates on large-font short snippets**: At 0.5 m, CET6 document pages yield 0.4–

TABLE 10. Real-Capture VLM Probes (0° On-Axis; All Rows Except the Original Short Baseline Are High-Suppression). Each VLM Cell Reports Exact Count / Char / *N* (Recoveries / % / Analyzed Calls). OCR BoE Shows the Same-Batch Three-Engine Best-of-Engine Reference (Exact / Char, %). The Temporal-Mean Rows Average 150 Short-Exposure Frames; Other Aggregate Views Are Defined in the Text.

Distance	Condition	OCR BoE	Qwen3-VL-32B	Kimi-K2.6	GLM-4.5V
Short-exposure distance sweep					
0.5 m	original short	58.3 / 92.4	30 / 94.5 / 36	30 / 95.6 / 36	19 / 92.0 / 26
0.5 m	short	0.0 / 8.4	28 / 91.5 / 36	17 / 84.7 / 36	16 / 70.8 / 31
1.0 m	short	0.0 / 3.6	20 / 79.1 / 36	14 / 68.6 / 36	9 / 47.0 / 33
1.5 m	short	0.0 / 6.5	17 / 68.9 / 36	12 / 61.5 / 36	13 / 70.0 / 36
Key attack views					
0.5 m	long	0.0 / 2.1	28 / 85.7 / 34	1 / 7.9 / 36	0 / 4.9 / 30
0.5 m	Video (150-frame mean)	8.3 / 50.7	10 / 94.0 / 12	10 / 95.5 / 12	7 / 91.8 / 8
1.5 m	long	0.0 / 13.2	33 / 89.3 / 36	32 / 89.9 / 36	21 / 62.4 / 36
1.5 m	Video (single best)	0.0 / 4.3	1 / 24.8 / 12	0 / 12.7 / 12	0 / 29.8 / 12
1.5 m	Video (150-frame mean)	0.0 / 48.4	8 / 90.3 / 12	8 / 90.3 / 12	7 / 88.5 / 12
1.5 m	Video (window-mean best)	0.0 / 8.6	7 / 89.0 / 12	5 / 84.2 / 12	8 / 87.2 / 12
1.5 m	Video (max projection)	0.0 / 19.0	8 / 90.1 / 12	7 / 89.2 / 12	7 / 88.7 / 12

TABLE 11. 0.5 m High-Suppression Short-Exposure VLM Character Recovery by Content Type. Baseline Column Shows Unprotected Original Short Results for Qwen (Kimi/GLM Similar); Parenthesized *N* Shows the Number of Analyzed Calls. At 1.5 m, CET6 Dense Pages Yield 0% Char Across All Three Models.

Content type (N)	Baseline (%)	Qwen (%)	Kimi (%)	GLM (%)
CET6 document (3)	64.4	18.5	5.6	0.4
Credentials (6)	97.6	97.6	88.3	77.1 (5)
Code snippets (6)	89.8	94.9	95.8	88.9
Digit strings (6)	100.0	100.0	98.6	50.0 (4)
Mixed content (6)	100.0	100.0	92.5	98.2 (4)
Paragraphs (6)	100.0	99.8	81.7	66.5
English sentences (3)	95.3	95.3	96.9	94.6

18.5% VLM character recovery, whereas credentials, code, and digit strings yield 50–100% (Table 11). At 1.5 m, CET6 pages drop to 0% across the three models, while digit strings and short sentences remain recoverable. The pattern is consistent with font scale and language priors [50], [51].

3) **Attack capability varies across models and conditions**: At 1.5 m, nominal long-exposure exact recovery ranges from 58.3% to 91.7%, with 62.4–89.9% character recovery. Endpoint, distance, capture view, and content jointly shape this spread.
4) **Sustained video aggregation strongly increases VLM recovery**: At 0.5 m, the 150-frame mean yields 83.3–87.5% exact and 91.8–95.5% character recovery across models. At 1.5 m, the 150-frame mean yields 58.3–66.7% exact and 88.5–90.3% character recovery, while single-best-frame exact recovery is 0–8.3%. The accompanying 12.7–29.8% single-best character values rest on low-information inputs, where fluent language-prior completions can inflate character recovery even though the prompt prohibits guessing; the exact-match counts are the more conservative reading.

E. USER EXPERIENCE STUDY

To evaluate the impact of the protected display on human readability and visual comfort, we designed and conducted a web-based user study.

Equipment and environment protocol: The same 240 Hz display as in §V-A. The formal study version sets a hard measured refresh-rate threshold of $\geq$200 Hz to ensure the fixed $n = 4$ basic subframe cycle rate $f_r/n \geq 50$ Hz; below this threshold participation is blocked rather than silently reducing n. The 144–199 Hz adaptive path is available only for an explicitly labeled demonstration mode, and its sessions are excluded from analysis by default. Before each participant begins, the experimenter confirms fixed display brightness, disabled auto-brightness/power-saving and variable refresh rate, unified browser fullscreen, closed high-load background processes, and an approximate 60 cm viewing distance. Refresh rate is re-measured after any display-mode change.

The browser records actual frame intervals in each trial's `requestAnimationFrame` loop; if the interval between adjacent callbacks exceeds 1.5× the expected interval computed from the pre-trial measured refresh rate, a dropped-frame event is logged, along with estimated drop count, drop rate, mean/max frame intervals, observed refresh rate, basic subframe cycle rate, and full cycle rate with inversion frame. This recording reflects only browser presentation cadence and cannot substitute for panel scanning or photometric measurement.

Ethics and safety: The study is minimal-risk human-subjects research. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type, so no institutional protocol number exists. Every session was administered in person by the experimenter under the safeguards described here. Before each session, the experimenter individually confirmed the participant's voluntary consent and photosensitivity-screening response in person. The web page then reiterated that the masked display involves rapid temporal modulation, that participants should stop immediately upon experiencing discomfort, eye fatigue, dizziness, nausea, or headache, and that they could withdraw unconditionally at any time. Proceeding recorded the confirmation timestamp and screening flag. Individuals who reported a history of photosensitive epilepsy or sensitivity to flicker were not eligible to continue. The implementation applies a conservative $f_r/n \geq 50$ Hz operating floor, but this engineering gate is not an ethical or clinical safety determination. At $n = 4$ on a 240 Hz panel the basic cycle rate is 60 Hz; the inversion frame reduces the full cycle frequency to $f_r/(n+1) = 48$ Hz, and the implementation separately logs warnings and actual timing. The formal study did not collect names, student identifiers, majors, ages, or gender information. Analysis and publication use only de-identified study records. Vision-correction status was collected to characterize the analyzed sample.

Task design: Each participant first completes one 12-second, unscored unmasked Canvas typing warm-up, views a 10-second, unscored readability-priority full-mask preview, and then completes an 8-second, unscored typing practice under the same readability-priority mask. This sequence separates initial stimulus and input adaptation from the four subsequent 20-second scored trials (within-subjects design). On every typing screen, including warm-up, practice, and scored trials, the stimulus Canvas is visible and its player has started before the participant clicks "Start." Clicking "Start" triggers a 5-second countdown while the input remains disabled; the timed typing interval and input availability begin only when the countdown ends. The same pre-start viewing procedure applies to both control and masked conditions.

The two conditions are each repeated twice. The experimenter's continuously assigned zero-based sequence number k selects ABBA or BAAB order to counterbalance practice and fatigue: (A) **original text** (control), displayed as unmasked Canvas with $n = 1$; (B) **masked text**, using the readability-priority configuration: $n = 4$, mask+noise, strong anti-OCR ($\alpha_s = 0.10$, $\alpha_g = 0.12$, 6 cycles) + weak inversion ($\alpha = 0.2$). The typing order index is $k \bmod 2$, and the subjective-rating Latin-square row is $\lfloor k/2 \rfloor \bmod 6$. Every consecutive 12 participants therefore cover all 2×6 joint assignments. Formal sessions enforce a uniqueness constraint on k, and the backend recomputes and verifies both order indices.

Both conditions are rendered by the same `renderSourceCanvas` path, with identical dark background, light text, canvas size, 23 px bold font, and color. Each control/masked stimulus is generated independently by the same seeded pseudo-word procedure with the same 220-character target length; this matches length and generative distribution, but does not make the literal strings identical. Temporal protection is the condition manipulation, while display order and stimulus generation remain potential task-level variation. Here "cycles" means that the web implementation rotates through six distinct mask, noise, and stripe instances before repeating. This web-playback-specific setting shares α_s and α_g with the readability-priority physical-capture configuration in §V-A and avoids repeatedly integrating one fixed overlay pattern. The web player is the independent JavaScript/Canvas2D path described in §IV: deterministic seeded masking, pixel-space gain, and surrogate texture noise, rather than the ChaCha20- and gradient-based Python pipeline. Frame-scheduling logs verify presentation timing at the JavaScript level; no photometric measurement of per-subframe panel output was performed.

Scoring no longer compares characters position-by-position but aligns the full transcription against the optimal target prefix using Levenshtein minimum string distance (MSD) [52]; untyped target suffix does not count as error. The alignment yields accuracy, characters per minute (CPM), and words per minute (WPM), along with edit distance, aligned prefix length, and scoring version. First-keystroke latency is defined from the end of the 5-second countdown, when the input is enabled and the scored timer starts, to the first input event that makes the transcription non-empty. Analysis first averages each participant's two trials per condition, then performs within-subject paired comparison.

Subjective ratings: After the typing trials, participants rate 6 conditions on the three 1–5 Likert items displayed by the interface: readability (1 = difficult to read, 5 = very clear), stability (stored in the `flicker` field; 1 = strong flicker, 5 = no perceptible flicker), and immediate visual comfort (`fatigue`; 1 = very uncomfortable, 5 = very comfortable). Higher values therefore indicate a better outcome on every item. Conditions include the unmasked original anchor, $n \in \{2, 3, 4\}$ mask+noise, $n = 4$ mask-only, and the same $n = 4$ readability-priority full configuration (mask+noise+strong anti-OCR+weak inversion) used in the typing task. The first three masking levels all achieve $\geq$60 Hz real temporal playback on a 240 Hz panel; the originally planned $n = 8$ would yield 30 Hz and trigger static fallback, so it is not included as a temporal ablation. The six conditions are presented in a 6th-order balanced Latin square; each condition must be viewed for at least 10 seconds before submission is enabled, with viewing start/end times and duration recorded. The unmasked anchor is also an attention/scale-interpretation aid but is not mechanically excluded by a "must rate 5" rule. This study does not include the high-suppression profile; assessment of its visible stripes and grain's acceptability is left to future work (see §VI).

Sample size and analysis plan: A minimum of $N = 24$ participants was planned, a number within the range required for paired effect sizes of $d_z \approx 0.6$–0.8 at 80% power, and exactly covering two rounds of the 12-person joint order assignment. Pre-specified exclusion criteria are debug/demo sessions, incomplete four typing or six rating records, measured refresh rate below 200 Hz, any typing trial with fewer than 5 attempted characters, control mean accuracy below 50%, masked trials not in temporal mode, any trial with observed basic cycle rate below 50 Hz, or minimum-view straight-line ratings. The last criterion is defined as all six rating rows having a recorded viewing duration no greater than 11 seconds (a 1-second tolerance above the 10-second submission gate) and, within every row, identical scores on all

TABLE 12. **Typing Task: Within-Subject Comparison of the Control and Readability-Priority Masked Conditions ($N = 56$; Participant Means Over Two Trials per Condition; ′ = Masked − Control With Participant-Level Bootstrap 95% CI). Per Metric, a Paired t-Test Is Used When the Pre-Specified Shapiro Test Passes (Effect Size Cohen's d_z), Otherwise a Wilcoxon Signed-Rank Test (Effect Size Rank-Biserial r_{rb}); Test Statistics Are Reported in the Text**

Metric	Control	Masked	Δ [95% CI]
Accuracy (%)	96.54	94.83	−1.71 [−3.39, −0.04]
WPM	20.44	19.30	−1.14 [−1.91, −0.39]
CPM	102.19	96.48	−5.71 [−9.56, −1.88]
Attempted chars	35.21	33.47	−1.74 [−2.88, −0.62]
First-key latency (ms)	742.64	833.61	+90.96 [−31.11, +210.69]

TABLE 13. **Subjective Ratings for the Six Display Conditions ($N = 56$; 1–5 Likert, Higher Is Better; Mean ± SD). Stability Is the Inverse-Flicker Item and Comfort the Immediate Visual Comfort Item (§V-E); Friedman and Holm-Corrected Pairwise Wilcoxon Statistics Are Reported in the Text**

Condition	Readability	Stability	Comfort
Unmasked anchor	4.80 ± 0.64	4.79 ± 0.68	4.75 ± 0.61
n=2 mask+noise	4.61 ± 0.68	4.77 ± 0.57	4.48 ± 0.87
n=3 mask+noise	4.21 ± 0.87	4.57 ± 0.76	4.11 ± 0.91
n=4 mask+noise	4.23 ± 0.91	4.38 ± 0.93	4.02 ± 1.05
n=4 mask-only	4.16 ± 0.87	4.45 ± 0.78	4.02 ± 0.96
Readability-priority full (n=4)	3.79 ± 1.17	2.98 ± 1.30	3.39 ± 1.25

three items. WPM, CPM, accuracy, attempted characters, and first-keystroke latency are first averaged within each participant by condition; speed metrics must be interpreted jointly with accuracy to avoid misinterpreting fast input under low accuracy as performance improvement. Paired differences are tested using paired t-tests when the pre-specified Shapiro test is passed, otherwise Wilcoxon signed-rank tests, with effect sizes and participant-level bootstrap 95% confidence intervals reported. Likert data are analyzed per dimension using Friedman tests with Holm-corrected post-hoc pairwise Wilcoxon tests. Analysis scripts generate de-identified participant means, JSON audit reports, and two LaTeX tables directly from the formal database `study_formal.db`.

Participants: 61 volunteers were recruited; 56 sessions met all pre-specified inclusion criteria and entered analysis (5 excluded: 3 for minimum-view straight-line ratings and 2 for control mean accuracy below 50%; the per-criterion audit accompanies the released analysis output). The 61 formal sessions filled each of the twelve joint order buckets five times, with one bucket receiving a sixth session. The formal study did not collect names, student identifiers, majors, ages, or gender information, so no age or gender distribution is reported; 46 of the 56 wore glasses, 1 wore contact lenses, and 9 reported no vision correction. All formal sessions ran on the 240 Hz laboratory display with a measured refresh rate of at least 200 Hz.

Results: Table 12 and Fig. 6 summarize the typing task. Relative to the control condition, the readability-priority full configuration changed mean accuracy by −1.71 percentage points (95% CI [−3.39, −0.04], Wilcoxon $W = 428$, $p = 0.043$, $r_{rb} = -0.33$), WPM by −1.14 (95%

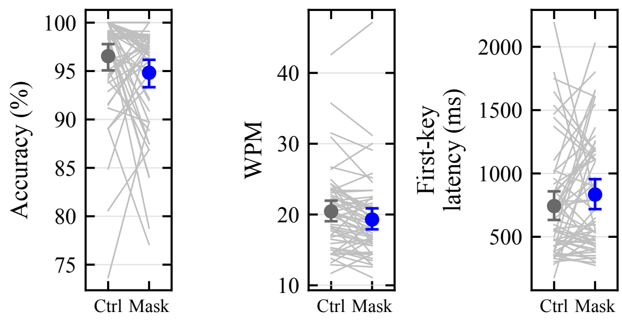

FIGURE 6. **Within-subject typing performance under the control and readability-priority masked conditions: each gray line is one participant (mean of two trials per condition); markers show group means with participant-level bootstrap 95% confidence intervals. Panels show transcription accuracy, words per minute, and first-keystroke latency.**

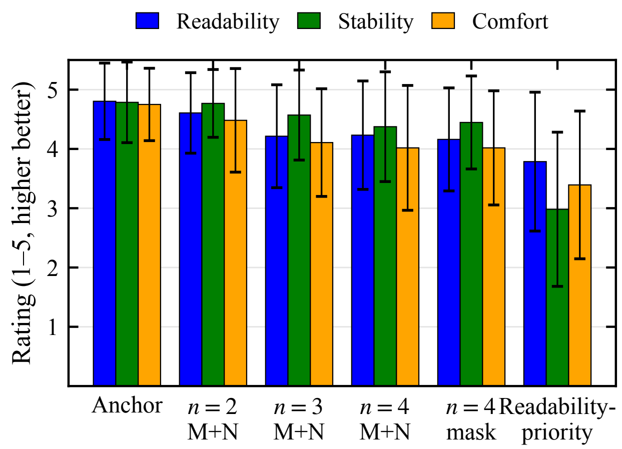

FIGURE 7. **Mean 1–5 Likert ratings (error bars: SD) for readability, stability, and immediate visual comfort across the six rated display conditions, ordered from the unmasked anchor to the readability-priority full configuration. Higher is better on every item.**

CI $[-1.91, -0.39]$, $t(55) = -2.93$, $p = 0.005$, $d_z = -0.39$), and first-keystroke latency by $+91.0$ ms (95% CI $[-31.1, +210.7]$, Wilcoxon $W = 641$, $p = 0.200$, $r_{rb} = 0.20$); CPM and attempted characters gave $p = 0.005$ ($d_z = -0.39$) and $p = 0.004$ ($d_z = -0.40$), respectively. Under Holm correction across the five typing metrics, WPM, CPM, and attempted characters remain significant whereas the accuracy difference ($p = 0.043$, rank 4 of 5) does not. Table 13 and Fig. 7 summarize the subjective ratings: Friedman tests yielded $\chi^2(5) = 72.2$ ($p = 3.5 \times 10^{-14}$) for readability, $\chi^2(5) = 123.2$ ($p = 6.5 \times 10^{-25}$) for stability, and $\chi^2(5) = 98.5$ ($p = 1.1 \times 10^{-19}$) for immediate visual comfort; Holm-corrected pairwise comparisons between the readability-priority full configuration and the unmasked anchor gave $p = 2.6 \times 10^{-4}$, $p = 2.2 \times 10^{-7}$, and $p = 1.8 \times 10^{-6}$ on the three items, respectively.

TABLE 14. Real-Capture COCO Detection (150 Images; `real_clean` = Unprotected Capture Baseline)

Detector	Condition	mAP	mAP50	AR
YOLO26x	`real_clean`	28.6%	43.9%	31.0%
	`real_short`	1.3%	2.4%	1.4%
	`real_video`	3.5%	5.3%	3.6%
RT-DETR-x	`real_clean`	30.1%	50.2%	34.8%
	`real_short`	3.5%	5.8%	4.1%
	`real_video`	4.6%	7.3%	5.5%
Faster R-CNN	`real_clean`	21.3%	39.2%	28.7%
	`real_short`	0.8%	1.6%	0.9%
	`real_video`	1.5%	3.0%	1.8%
RetinaNet	`real_clean`	22.6%	39.4%	27.0%
	`real_short`	0.9%	1.8%	1.0%
	`real_video`	1.9%	3.1%	2.0%

TABLE 15. Real-Capture MOT17-09-FRCNN Still-Frame Tracking (ByteTrack-Style Greedy Association, 450 Frames; HOTA and IDF1 Computed by TrackEval [58])

Detector	Condition	HOTA	IDF1
YOLO26x	`real_clean`	15.8%	12.4%
	`real_short`	8.7%	6.5%
	`real_video`	10.0%	8.7%
RT-DETR-x	`real_clean`	14.2%	10.5%
	`real_short`	8.1%	5.4%
	`real_video`	9.6%	6.9%
Faster R-CNN	`real_clean`	14.4%	9.8%
	`real_short`	6.8%	6.2%
	`real_video`	10.5%	7.6%
RetinaNet	`real_clean`	14.2%	10.3%
	`real_short`	5.1%	4.4%
	`real_video`	6.8%	5.9%

F. REAL-CAPTURE OBJECT DETECTION AND TRACKING

To probe whether temporal degradation transfers beyond text, we report an exploratory detection/tracking stress test within the same display–camera link; its evidence strength is not equivalent to the matched OCR main experiment. Content was displayed on the 240 Hz screen, captured by the S600 at 1.5 m / 0°, and processed by four detectors [53]–[56]. mAP, mAP50, and average recall (AR) are reported for detection. Real-capture tracking uses higher-order tracking accuracy (HOTA) and identification F1 (IDF1); simulated tracking additionally reports multiple-object tracking accuracy (MOTA) and multiple-object tracking precision (MOTP). The protected condition uses the $n = 4$ mask+noise profile without overlays or inversion. The `real_clean` reference is an unprotected nominal 31.25-ms capture. The `real_short` image is the maximum-grayscale-contrast frame from a three-frame protected burst at the nominal 3.91-ms setting, and `real_video` is the mean of 150 protected frames acquired nominally at 60 fps. Tracking uses 450 frames from MOT17-09-FRCNN via still-display-then-capture and an in-project greedy association implementation inspired by ByteTrack [57]; it does not establish performance for a continuous-video tracker.

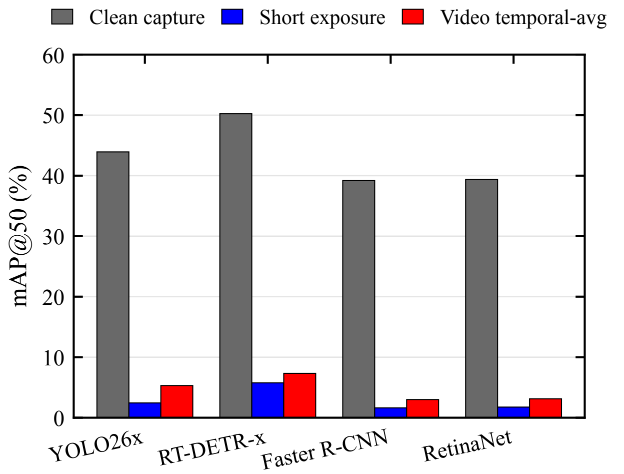

FIGURE 8. Real-capture mAP50 for four detectors across the unprotected capture, short-exposure protected capture, and 150-frame protected mean. Conditions differ in protection, exposure, resolution, and frame reduction.

Under short exposure, all four detectors' mAP50 drops from 39.2–50.2% to 1.6–5.8% (Table 14, Fig. 8), representing relative decreases exceeding 85%, while the `real_video` condition yields 3.0–7.3%, also below the `real_clean` baselines. Because protection, exposure, resolution, and frame reduction differ across conditions, these values are a cross-task stress test rather than an isolated protection effect. In still-frame tracking (Table 15), HOTA decreases from 14.2–15.8% to 5.1–8.7% and IDF1 from 9.8–12.4% to 4.4–6.5%. Detection mAP was computed by pycocotools. HOTA and IDF1 were both computed by TrackEval. The tracker is an in-project greedy IoU association implementation.

G. SOFTWARE SIMULATION EXPERIMENTS

This section uses software simulation to address two questions: (1) what is the machine readability of one protected subframe obtained after the application renderer, and (2) what is recovery under zero camera noise and perfect alignment? By characterizing the post-render single-subframe capture point, the experiment defines a boundary that malware intercepting the original frame or accumulating a complete cycle would bypass.

1) Synthetic Single-Subframe OCR

Using Tesseract [46], EasyOCR [45], and Surya [44] on 120 synthetic samples, character recovery drops from 94.0–97.0% to 0–2.6% across three engines (Table 16). Stratified breakdowns further characterize recovery across the evaluated content categories.

Fig. 9 visualizes these results; single-subframe recovery does not exceed 2.6% for any engine.

TABLE 16. Synthetic Single-Subframe OCR: Three-Engine Character Recovery (120 Samples, $n = 4$, $\varepsilon = 8/255$)

OCR engine	Original frame	Single subframe	Reduction
Tesseract	94.0%	0.0%	94.0%
EasyOCR	94.1%	0.0%	94.1%
Surya	97.0%	2.6%	94.4%

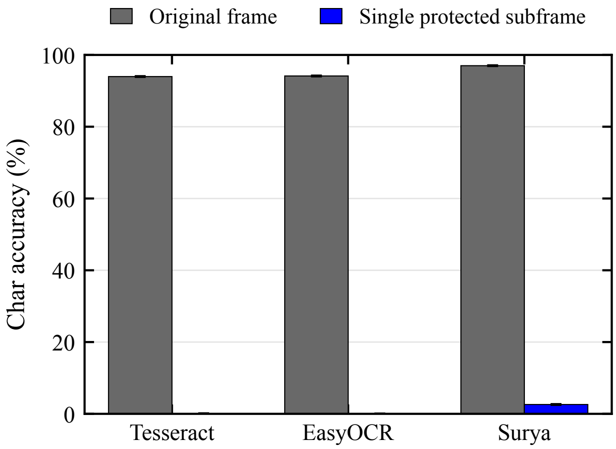

FIGURE 9. Character recovery for three OCR engines on original frames and single protected subframes; error bars are 95% bootstrap intervals resampled by corpus sample.

TABLE 17. Archived Digital Reconstruction Boundary (Tesseract)

Setting	N	Char.	Exact	SSIM	ΔE_{00}
Base: cycle mean	120	94.3%	85.8%	–	–
Base: per-sample strongest	120	95.2%	91.7%	–	–
Readability: cycle mean	16	87.9%	50.0%	0.948	1.46
High: cycle mean	16	56.6%	6.3%	0.764	4.87
High: inversion slot	16	93.0%	81.2%	–	–

The 120-sample base attack and 16-sample profile ablation are separate archived digital evaluations. "Per-sample strongest" selects the best result among seven prespecified reconstructions (temporal average cycle, blue channel max, differential blue, differential luma, global shutter slot 0, phase search max, and phase search mean) for each sample. "Inversion slot" extracts only the $\alpha(\mathbf{1} - \mathbf{I})$ frame and applies contrast stretching; its 93.0% recovery identifies slot-level composition as a critical design variable. Profile SSIM and ΔE_{00} describe full-cycle reconstruction quality, not physical or user-perceived quality.

2) Theoretical Security Boundary

When an attacker accumulates registered frames covering a full cycle, complementary subframes can be re-summed in the ideal linear model of (1), with the archived per-sample strongest digital attack recovering 95.2% and pure full-cycle temporal averaging recovering 94.3% (Table 17)—values that empirically mark the reconstruction boundary of the evaluated linear decomposition. Across the nine physical geometries, the high-suppression 150-frame mean yields 47.9% recovery. Character recovery also depends on content and the physical imaging path.

TABLE 18. COCO val2017 Simulated Detection Pipeline Diagnostics (4 Detectors, 8 Images)

Detector	Condition	mAP	mAP50	AR
YOLO26x	clean	48.1%	58.3%	51.0%
	single subframe	0.0%	0.0%	0.0%
	temporal avg.	14.3%	18.3%	14.3%
RT-DETR-x	clean	47.1%	58.7%	53.4%
	single subframe	5.2%	6.0%	5.2%
	temporal avg.	16.8%	23.7%	17.9%
Faster R-CNN	clean	38.2%	52.7%	43.6%
	single subframe	3.0%	4.4%	3.0%
	temporal avg.	11.1%	16.5%	11.1%
RetinaNet	clean	33.2%	47.1%	35.3%
	single subframe	3.3%	4.2%	3.3%
	temporal avg.	9.6%	13.2%	9.6%

TABLE 19. MOT17 Simulated Tracking (ByteTrack-Style Greedy Association; 5,316 Frames; CLEAR and IDF1 by In-Project Approximate Implementation)

Detector	Condition	MOTA	MOTP	IDF1
YOLO26x	clean	31.1%	81.7%	27.1%
	single subframe	0.1%	74.8%	0.1%
	temporal avg.	9.1%	78.1%	12.0%
RT-DETR-x	clean	24.1%	79.8%	38.2%
	single subframe	−2.3%	73.2%	5.4%
	temporal avg.	−11.4%	75.0%	14.7%
Faster R-CNN	clean	3.8%	77.9%	35.6%
	single subframe	−2.5%	71.2%	4.5%
	temporal avg.	−2.7%	73.2%	14.0%
RetinaNet	clean	−5.4%	77.1%	30.5%
	single subframe	0.4%	70.6%	2.7%
	temporal avg.	−2.9%	73.8%	11.9%

3) Detection and Tracking Simulation

Simulations were run on COCO val2017 [59] and MOT17 [60] with YOLO26x [53], RT-DETR-x [54], Faster R-CNN [55], and RetinaNet [56]: the COCO simulation used 8 images as an implementation pipeline diagnostic (Table 18), while the MOT simulation covers 7 sequences totaling 5,316 frames and uses the same in-project ByteTrack-inspired greedy association method (Table 19).

On the 8 COCO images, single-subframe mAP50 across the four detectors is 0–6.0%. Across the 5,316 MOT frames, single-subframe IDF1 is 0.1–5.4%, while MOTA includes negative and nonmonotonic values. Negative MOTA occurs when the false-positive, false-negative, and identity-error penalties exceed the number of ground-truth objects; it is not a negative detection probability. Detector architecture, confidence thresholds, proposal density, and the greedy associator jointly shape the observed differences. Fig. 10 summarizes the residual capability of a single protected subframe relative to the clean baseline across the evaluated OCR, detection, and tracking pipelines.

VI. DISCUSSION

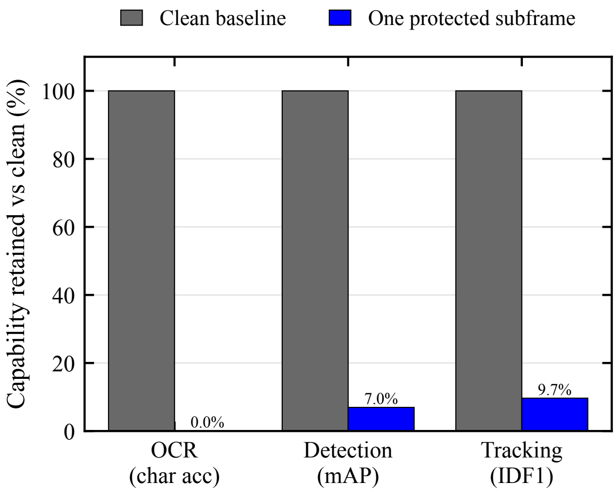

FIGURE 10. Residual capability of one protected subframe relative to the clean baseline (clean = 100%) across the evaluated OCR, detection, and tracking pipelines. The metrics are normalized within task and are not directly commensurate across tasks.

A. PRINCIPAL FINDINGS AND RESEARCH VALUE

The primary result is a matched short-exposure comparison in the tested S600 display–camera link. Across 12 content items, 9 geometries, and 3 repeats, best-of-engine character recovery falls from 94.1% for unprotected content to 16.2% for readability-priority and 5.0% for high-suppression. The corresponding content-cluster contrasts are 78.0 and 89.1 percentage points. Thus, the result establishes feasibility for mitigating conventional OCR under the evaluated short-exposure conditions, rather than a general guarantee against screen photography.

The research value lies in treating protection as a system property rather than inferring it from digitally generated subframes alone. The evaluation connects multi-engine OCR, explicit-field leakage and upper-tail risk, fixed preprocessing, temporal integration, digital reconstruction, and VLM probes within the same display–camera setting. Together, these analyses characterize not only when conventional OCR recovery decreases, but also how recovery changes as the attack capability is strengthened. Detection and tracking remain secondary cross-task stress tests and do not carry the same evidentiary weight as the matched OCR analysis.

B. INTEGRATION AND ATTACKER ESCALATION

Temporal masking exploits the difference between human temporal integration and short camera exposure; in the ideal linear model, registered complementary subframes can reconstruct the original because $\sum \mathbf{I}_k = \mathbf{I}$. The digital attack evaluation makes this boundary explicit: per-sample strongest reconstruction reaches 95.2% recovery and pure full-cycle averaging reaches 94.3%. These values characterize the reconstruction boundary of the evaluated digital decomposition; they are not a measure of physical display performance, human readability, or confidentiality in the camera link.

The physical temporal-averaging attack likewise recovers substantial text without the mask seed or phase. Across the nine geometries, the 150-frame mean reaches 71.1% recovery for readability-priority and 47.9% for high-suppression. Sustained capture can therefore convert temporal fragments into a usable recognition signal, while the extent of recovery remains coupled to the evaluated imaging conditions.

C. PROFILE TRADE-OFFS AND PRACTICAL IMPLICATIONS

The two evaluated profiles exhibit distinct operating points. Readability-priority has a digitally reconstructed SSIM of 0.948 but retains 15.7% field-micro exact recovery across the nine-geometry matched pool, whereas high-suppression lowers several recovery measures while reducing SSIM to 0.764 and increasing ΔE_{00} from 1.46 to 4.87. These SSIM and color-difference values describe full-cycle digital reconstructions, not human readability, comfort, or physical panel luminance; they therefore identify a digital reconstruction–recovery trade-off rather than a continuously validated tuning curve.

The user study supplies the human-facing evidence for the readability-priority operating point. In the independent JavaScript/Canvas2D implementation and short typing task, participants retained 94.83% typing accuracy but typed 1.14 WPM more slowly; readability, stability, and comfort averaged 3.79, 2.98, and 3.39 out of 5. These results make the speed and perceived-stability costs explicit, but they do not substitute for physical photometric measurement of the capture pipeline.

This objective differs from Kaleido-class temporal re-encoding in its treatment of completeness [10]. The present composite profiles intentionally depart from strict completeness because full-cycle completeness creates an entry point for integration attacks on static text. At the same time, 93.0% recovery from the high-suppression inversion slot after contrast stretching identifies slot-level composition as a weakness in the current digital attack evaluation. Temporal masking should therefore be treated as one display-layer control that can be paired with field-level policies and independent access controls for high-value content, not as a standalone protection guarantee.

D. VLM-AWARE IMPLICATIONS

The VLM probes show that conventional OCR is not an adequate proxy for all recognition attackers. On the protected photographs, recovery is strongest for large-font credentials, code, digits, and short sentences, while dense CET6 pages are more strongly suppressed. These results show that high-suppression can reduce conventional OCR recovery while stronger recognition still recovers substantial large-font content.

E. LIMITATIONS AND FUTURE WORK

The physical evidence is limited to one S600 UVC webcam, and profiles were captured in a fixed rather than randomized order. Exposure values are control-derived labels rather than photometric shutter measurements. The archive does not decode or independently measure auto-exposure state, gain, white balance, focus, ambient illuminance, display–camera phase, or physical panel luminance. These factors can interact with time, geometry, and profile order, which limits both causal isolation and external generalization.

The statistical and attack evidence is also bounded. Content-cluster intervals hold the tested geometries fixed, and the 12-item corpus is too small for script-level inference. The 150-frame experiment does not map the duration–recovery curve. Detection uses 150 COCO images at one position, while tracking uses 450 still-display-then-capture frames from one MOT17 sequence. Commercial OCR, learned restoration, and unconstrained adaptive preprocessing may recover additional text.

The current PoC operates at the application layer and lacks photometric slot-timing measurements, luminance-matched static controls, and matched noise-off versions of the enhanced profiles. Temporal sparsity, duty-cycle luminance, overlays, panel response, camera exposure, and image signal processor processing therefore remain coupled. Rolling-shutter row mixing and panel transitions are candidate explanations for the simulation-to-real gap. The user evidence comes from an independent browser implementation and a short task, and it does not assess the visibly stronger high-suppression profile.

Future work should randomize profile order, lock camera parameters, record batch time, and measure physical slot timing and luminance. Matched static controls can separate temporal and luminance effects. Replication across cameras and displays can test transfer, while continuous-video attacks can map recovery against recording duration. Longer usability studies across panels, including the high-suppression profile, can further characterize the observed security–usability tradeoff.

VII. CONCLUSION

We evaluated temporal pixel masking on one UVC camera link. Across 9 geometries, the same 324 matched units per profile (12 content items $\times$ 9 geometries $\times$ 3 repeats) yield 94.1% mean recovery for unprotected content, 16.2% for readability-priority, and 5.0% for high-suppression. The corresponding paired content-cluster contrasts are 78.0 and 89.1 percentage points. These results show that conventional OCR recovery can be substantially reduced in the evaluated short-exposure display–camera link; they do not establish a general guarantee against screen photography.

The attack-oriented evaluation also defines the limits of that result. Fixed preprocessing, full-cycle digital reconstruction, temporal integration, and VLM probes recover substantial information under their respective conditions. Temporal pixel masking is therefore a conditional mitigation against conventional short-exposure OCR, not a general defense against sustained recording, registered reconstruction, or stronger recognition systems.

In a 56-participant user study, readability-priority masking retained high typing accuracy while imposing measurable speed and perceived-stability costs. Detection and tracking provide secondary cross-task stress tests under coupled display, capture, and reduction conditions; they are not substitutes for the matched OCR evidence. Together, the results position the contribution as a system-level empirical characterization of both the feasible operating region and the attack boundary of temporal pixel masking.

ETHICS STATEMENT

This research is defensive in nature and evaluates recovery boundaries to assess protection against unauthorized screen photography, not to provide an attack guide. The user study involved minimal-risk behavioral research. The authors' institution does not maintain an ethics review committee for non-medical behavioral experiments of this type, so no institutional determination was available. Before each session, the experimenter individually confirmed voluntary consent and photosensitivity-screening eligibility in person; the web interface recorded the confirmation state and timestamp. The protocol included in-person supervision and voluntary withdrawal as described in §V-E, and analysis and publication use only de-identified data.

DATA AVAILABILITY

The data and code that support the findings of this study are available from the corresponding author upon reasonable request.

ACKNOWLEDGMENT

[PLACEHOLDER: Acknowledgment to be filled.]

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